METRIC
DOD-B-53061
5 December 1985

# MILITARY SPECIFICATION

# BELTS, V: POLYURETHANE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers polyurethane, V-type, power transmission belts.

1.2 Classification. The belts shall be of the nominal top widths and thicknesses for the sizes indicated in table I, and of the effective lengths indicated in table II, as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3030

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TABLE I. Belt sizes.

| Size designation | Nominal top width mm[1/] | Nominal top width in.[2/] | Nominal thickness mm | Nominal thickness in. |
|---|---|---|---|---|
| 1 | 3.00 | 0.119 | 2.39 | 0.094 |
| 2 | 5.00 | 0.197 | 3.00 | 0.119 |
| 3 | 7.00 | 0.275 | 5.57 | 0.219 |
| 4 | 11.00 | 0.434 | 7.14 | 0.281 |

1/ millimeters

2/ inch

TABLE II. Belt effective lengths.

| Belt effective length | | Belt effective length | | Belt effective length | |
|---|---|---|---|---|---|
| mm | in. | mm | in. | mm | in. |
| 180[1/] | 7.09 | 315 | 12.40 | 560 | 22.05 |
| 185 | 7.28 | 325 | 12.80 | 580 | 22.83 |
| 190 | 7.48 | 335 | 13.19 | 600 | 23.62 |
| 195 | 7.68 | 345 | 13.58 | 615 | 24.21 |
| 200 | 7.87 | 355 | 13.98 | 630 | 24.80 |
| 206 | 8.11 | 365 | 14.37 | 650 | 25.59 |
| 212 | 8.35 | 375 | 14.76 | 670 | 26.38 |
| 218 | 8.58 | 387 | 15.24 | 690 | 27.17 |
| 224 | 8.82 | 400 | 15.75 | 710[4/] | 27.95 |
| 230 | 9.06 | 412 | 16.22 | 730 | 28.74 |
| 236 | 9.29 | 425 | 16.73 | 750 | 29.53 |
| 243 | 9.57 | 437 | 17.20 | 775 | 30.51 |
| 250 | 9.84 | 450 | 17.72 | 800 | 31.50 |
| 258 | 10.16 | 462 | 18.19 | 825 | 32.48 |
| 265 | 10.43 | 475 | 18.70 | 850 | 33.46 |
| 272 | 10.71 | 487 | 19.17 | 875 | 34.45 |
| 280[2/] | 11.02 | 500[3/] | 19.69 | 900 | 35.43 |
| 290 | 11.42 | 515 | 20.28 | 925 | 36.42 |
| 300 | 11.81 | 530 | 20.87 | 950 | 37.40 |
| 307 | 12.09 | 545 | 21.46 | 975 | 38.39 |



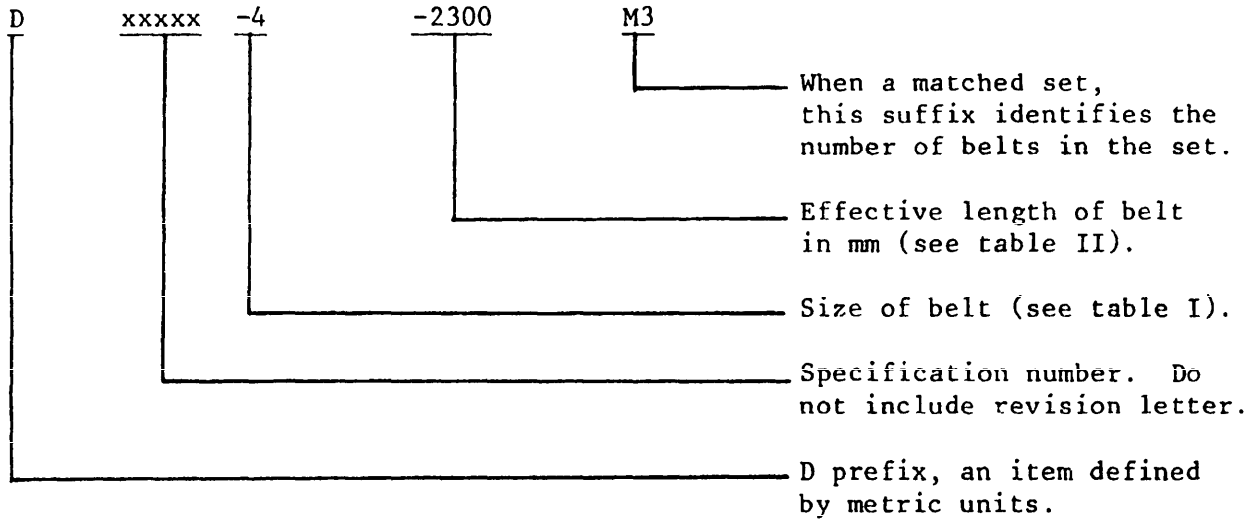
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TABLE II. Belt effective lengths. - Continued

| Belt effective length | | Belt effective length | | Belt effective length | |
|---|---|---|---|---|---|
| mm | in. | mm | in. | mm | in. |
| 1000 | 39.37 | 1320 | 51.97 | 1800 | 70.87 |
| 1030 | 40.55 | 1360 | 53.54 | 1850 | 72.83 |
| 1060 | 41.73 | 1400 | 55.12 | 1900 | 74.80 |
| 1090 | 42.91 | 1450 | 57.09 | 1950 | 76.77 |
| 1120 | 44.09 | 1500 | 59.06 | 2000 | 78.74 |
| 1150 | 45.28 | 1550 | 61.02 | 2060 | 81.10 |
| 1180 | 46.46 | 1600 | 62.99 | 2120 | 83.46 |
| 1220 | 48.03 | 1650 | 64.96 | 2180 | 85.83 |
| 1250 | 49.21 | 1700 | 66.93 | 2240 | 88.19 |
| 1280 | 50.39 | 1750 | 68.90 | 2300 | 90.55 |

1/ Size 1 belts in this length up to 750 mm (29.53 in.) length
2/ Size 2 belts in this length up to 1500 mm (59.06 in.) length
3/ Size 3 belts in this length up to 2300 mm (90.55 in.) length
4/ Size 4 belts in this length up to 2300 mm (90.55 in.) length

1.2.1 Part number. The military specification part numbers for the belts shall be as follows:




Example of part number: "Dxxxxx -4-2300M3" for a matched set of 3 belts, size 4, with an effective length of 2300 mm.



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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | - Boxes, Wood, Cleated Plywood. |
| PPP-B-621 | - Box, Wood, Nailed and Lock-corner. |
| PPP-B-636 | - Box, Shipping, Fiberboard. |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of U.S. Military Property. |

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be of the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 2137 - Rubber Property-Brittleness Point of Flexible Polymers and Coated Fabrics.
ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The polyurethane V-belts, hereinafter referred to as "belt", shall be an endless loop having an inward tapering cross-section.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all of the provisions of this specification.

3.3.1 Materiel deterioration prevention and control. The belt shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the belt may be exposed.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available to the contracting officer or designated representative.

3.3.3 Base material. The base material of the belt shall be polyurethane. Polyurethane material shall retain not less than 60 percent of its original tensile strength after immersion in water for 14 days at 82° C (180° F). Polyurethane material shall not be brittle when tested as specified in 4.5.2.2.

3.3.4 Cords. Cords shall be polyester impregnated with polyurethane. The polyurethane used to impregnate the cords shall conform to 3.3.3.

3.3.5 Fabric. Fabric, if used in the belt, shall be cotton impregnated with polyurethane.

3.3.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The belt may be newly manufactured from recovered materials to the maximum extent practicable, provided the belt produced meets all other requirements of this specification.



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3.4 Design. The inner circumference of the belt shall be a 60°, trapezoidal cross-sectioned compression member, surrounded by a rectangular rib containing imbedded tension members.

3.5 Construction. The belt shall be an integral composite of the members specified in 3.5.1 through 3.5.3.

3.5.1 Compression member. The inner compression member shall be all polyurethane molded into an endless loop of the appropriate sized cross-section. The compression member shall have a symmetrical V-profile and shall contain no chunked-out or melted spots.

3.5.2 Tension members. Tension members shall be cord or fabric uniformly spaced around the outer perimeter of the compression member in a direction parallel to the circumference of the belt. Tension members shall be neither frayed nor knotted. The application of the tension members around the compression member shall assure complete imbedment in the rib.

3.5.3 Rib. The rib shall be a rectangular polyurethane cushion around the outer perimeter of the compression member that surrounds and protects the tension members inbedded within. The rib shall be securely bonded to the compression member. The rib shall be fully formed and shall have no air bubbles, unfilled areas over the tension members, or chipped edges.

3.6 Performance. The belt shall perform as specified in 3.6.1 and 3.6.2.

3.6.1 Size 1 and 2 belts. When tested as specified in 4.5.2.3.2.2, the size 1 and 2 belts shall not turn over in the test pulleys, break, or exhibit evidence of fatigue.

3.6.2 Size 3 and 4 belts. When tested as specified in 4.5.2.3.2.5, the size 3 and 4 belts shall not turn over in the test pulleys, break, exhibit evidence of fatigue, or slip more than 8 percent.

3.7 Dimensions. The belt shall be of the size and effective length specified in table I and II. The belt shall conform to the rideout and effective length tolerances specified in 3.7.1 and 3.7.2.

3.7.1 Rideout. The top width, trapezoidal angle and thickness of the belt shall be such that the rideout of the belt shall not exceed the tolerances indicated in table III, when tested as specified in 4.5.2.3.1.



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The contractor shall design inspection equipment

TABLE III. Rideout and tolerances.

| Belt size | Nominal rideout | | Rideout tolerances (plus or minus) | |
|---|---|---|---|---|
| | mm | in. | mm | in. |
| 1 | 0.635 | 0.025 | 0.508 | 0.020 |
| 2 | 0.660 | 0.026 | 0.508 | 0.020 |
| 3 | 0.889 | 0.035 | 0.787 | 0.031 |
| 4 | 0.813 | 0.032 | 0.787 | 0.031 |

3.7.2 Effective length tolerances. When tested as specified in 4.5.2.3.1, the belt effective length shall not exceed the tolerances indicated in table IV.

TABLE IV. Belt effective length tolerance.

| Belt effective length range | | | | Center distance between test pulley centers tolerance (plus or minus) | | Overall belt length tolerance (plus or minus) | |
|---|---|---|---|---|---|---|---|
| from | | to | | | | | |
| mm | in. | mm | in. | mm | in. | mm | in. |
| 180 | 7.09 | 300 | 11.81 | 1.27 | 0.050 | 2.54 | 0.100 |
| 370 | 12.09 | 710 | 27.95 | 1.91 | 0.075 | 3.81 | 0.150 |
| 730 | 28.74 | 1090 | 42.91 | 2.54 | 0.100 | 5.08 | 0.200 |
| 1120 | 44.09 | 1500 | 59.06 | 3.18 | 0.125 | 6.35 | 0.250 |
| 1550 | 61.02 | 1900 | 74.80 | 3.81 | 0.150 | 8.13 | 0.320 |
| 1950 | 76.77 | 2300 | 90.55 | 4.45 | 0.175 | 8.89 | 0.350 |

3.8 Identification marking. The belt shall be identified in accordance with MIL-STD-130.

3.8.1 Single belt marking. Single belts shall have the identification applied to a paper tag tied to the belt. The part number shall be the identifying number. The date of manufacture by month and year, i.e., "January 1985" shall be marked on the tag.



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3.8.2 Matched set marking. Each belt in a matched set shall be marked as specified in 3.8.1. In addition, all belts of a matched set shall be tied together and a paper tag marked "WARNING - MATCHED SET - DO NOT SEPARATE PRIOR TO INSTALLATION" shall be tied to the set.

3.9 Alternate requirements. Alternate requirements for belts shall be as specified in 3.9.1 and 3.9.2.

3.9.1 Belt age. When specified (see 6.2), the age of the belt shall be not more than 12 months from the date of manufacture.

3.9.2 Matched sets. When specified (see 6.2), belts shall be furnished in matched sets of the quantity specified. All belts in a matched set shall conform to the effective length tolerances specified in table IV, except that the difference in effective length as measured between test pulley centers when tested as specified in 4.5.2.3.1, shall not exceed the matched set tolerances indicated in table V.

TABLE V. Matched set tolerances.

| Belt set effective length range | | | | Distance between test pulley centers tolerance (plus or minus) | | Deviation of effective length among all belts in set (plus or minus) | |
|---|---|---|---|---|---|---|---|
| from | | to | | | | | |
| mm | in. | mm | in. | mm | in. | mm | in. |
| 180 | 7.09 | 500 | 19.69 | 0.25 | 0.010 | 0.51 | 0.020 |
| 515 | 20.28 | 1000 | 39.37 | 0.51 | 0.020 | 1.02 | 0.040 |
| 1030 | 40.55 | 1500 | 59.06 | 0.76 | 0.030 | 1.52 | 0.060 |
| 1550 | 61.02 | 2300 | 90.55 | 1.02 | 0.040 | 2.03 | 0.080 |

3.10 Workmanship. The belt shall be fully formed and symmetical. When layed on a flat surface, the belt shall exhibit no evidence of twisting. The belt shall exhibit no evidence of delamination between rib and compression member, chatter marks or melting. The outer circumference of the belt shall be square with the inner circumference. The belt shall be free from imbedded foreign material and shall be clean on the exterior.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and facilities. The contractor shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2, in the order shown. Failure of any test shall be cause for rejection.



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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The belt shall be examined as specified herein for the following defects:

| Defect number | Defect | Requirement paragraph |
|---|---|---|
| Major | | |
| 101. | Material not inherently corrosion and deterioration resistant or not treated to provide protection against corrosion or deterioration. | 3.3.1 |
| 102. | Identification of material and finishes not available. | 3.3.2 |
| 103. | Base material not all polyurethane. | 3.3.3 |
| 104. | Cords not polyester impregnated with polyurethane. | 3.3.5 |
| 105. | Fabric, if used, not cotton impregnated with polyurethane. | 3.3.5 |
| 106. | Any material other than virgin or recovered material used in belt. | 3.3.6 |
| 107. | Design of belt not as specified. | 3.4 |
| 108. | Compression member not of symmetrical profile or contains chunked-out or melted spots. | 3.5.1 |
| 109. | Tension members frayed, knotted or not completely imbedded within rib. | 3.5.2 |
| 110. | Rib not securely bonded to compression member, or contains air bubbles, unfilled areas or chipped edges. | 3.5.3 |
| 111. | Age of belt more than 12 months, when age limit is specified. | 3.9.1 |
| 112. | Matched set not of quantity of belts specified. | 3.9.2 |
| 113. | Workmanship not as specified. | 3.10 |



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Minor

| | | |
|---|---|---|
| 201. | Identification marking not in accordance with MIL-STD-130. | 3.8 |
| 202. | Single belts, alone or in matched sets, not marked with part number and date of manufacture marked on tied tag. | 3.8.1, 3.8.2 |
| 203. | Matched sets not tied together and not tagged with warning notice. | 3.9.2 |

4.5.2 Tests.

4.5.2.1 Polyurethane tensile strength test. Test a specimen of polyurethane, from the same batch of polyurethane used in the manufacture of the belt, for tensile strength and record the tension at failure of the specimen. Immerse a like specimen from the same batch in distilled water maintained at not less than 82° C (180° F) for not less than 14 days. After 14 days, remove the immersed specimen from the water and subject it to the identical tensile strength procedure as the original specimen. Failure of the immersed specimen to record a tensile strength of 60 percent or more than the original specimen shall constitute failure of this test.

4.5.2.2 Polyurethane brittleness test. Subject a specimen of polyurethane, from the same batch of polyurethane used to manufacture the belt, to the test specified in ASTM D 2137 for 46 hours at minus 48° C (-54° F). Brittleness in the specimen at the end of the test period shall constitute failure of this test.

4.5.2.3 Test conditions, dimensional and performance tests. The tests for dimensions and performance specified herein shall be conducted in a ambient temperature of not less than 21° C (69.8° F) or more than 32° C (89.6° F).

4.5.2.3.1 Dimensional tests. The belt shall be tested for conformance to rideout and effective length as specified in 4.5.2.3.1.1 through 4.5.2.3.1.4.

4.5.2.3.1.1 Dimensional test fixture. The dimensional test fixture shall consist of a fixed pulley spindle, a horizontal moveable pulley spindle, a fixed scale, vertical pointers on both pulley spindle centers, a means of tensioning the moveable spindle, a means of locking the moveable pulley spindle in tension and a set of test pulleys. The dimensional test fixture configuration shall be as illustrated in figure 1. Test pulley dimensions shall be as shown in the figure.



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4.5.2.3.1.2 Test procedure. Mount applicable size test pulleys on the spindles for the size of test belt. Place the test belt on test pulleys and apply applicable tension as shown on the figure for the size of test belt. Lock-in the tension. Rotate the test belt two complete revolutions to seat belt firmly in pulley grooves and divide the tension equally on the upper and lower test belt strands.

4.5.2.3.1.3 Rideout test measurement. Place a straightedge across the top of the test belt. Measure the distance from the bottom of the straightedge to the outer rim of the test pulley. A rideout measurement that is not within the rideout tolerances specified for the size of belt being tested as shown in table III, shall constitute failure of this test.

4.5.2.3.1.4 Effective length test measurements. Record the center distance shown by the two pointers on the scale. Obtain overall effective length by multiplying the center distance recorded by two, and adding a factor of 3 inches for size 1 belts and 10 inches for size 2, 3 and 4 belts. An effective length measurement that is not within the tolerances specified in table IV, for the belt effective length range of the test belt, shall constitute failure of this test.

4.5.2.3.2 Performance tests. The belt shall be tested for performance as specified in 4.5.2.3.2.1 through 4.5.2.3.2.3 for size 1 and 2 belts, and as specified in 4.5.2.3.2.4 through 4.5.2.3.2.6 for size 3 and 4 belts.

4.5.2.3.2.1 Size 1 and 2 belt performance test fixture. The size 1 and 2 belt performance test fixture shall consist of: a fixed pulley spindle driven by a constant speed motor at 10,500 revolutions per minute (rpm), ±150 rpm; a vertical moveable pulley spindle; a means of tensioning the moveable spindle; a means of locking the moveable spindle in tension; a means of aligning both fixed and moveable spindles in the same true plane; and, a set of test pulleys. The size 1 and 2 performance test fixture configuration shall be as illustrated in figure 2. Test pulley sizes shall be as shown in the figure.

4.5.2.3.2.2 Size 1 and 2 belt performance test procedure. Mount the applicable size test pulleys on the spindles for the size of test belt. Place the test belt on test pulleys and apply the applicable tension shown in the figure for the test belt size. Lock-in the tension. Align the test pulleys to run in the same plane. Turn the constant speed drive motor on, and run the belt in the fixture for 5 minutes (min), to precondition the test belt. If the test belt breaks or turns over in the test pulleys during preconditioning, it shall constitute failure. Allow the belt to cool for not less than 10 min, then retension and realign, if necessary. Turn the drive motor on again and run the fixture for not less than 100 hours before stopping the drive motor. Release the tension, remove the belt and examine for evidence of fatigue.



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4.5.2.3.2.3 Size 1 and 2 belt performance test failure. If the test belt breaks or turns over in the pulleys at any time during the 100 hours run of the test fixture, it shall constitute failure of the test. If the belt exhibits evidence of fatigue, such as separation between rib and compression member in any degree, chipping of the rib, visible separation of a tension member, or chunking-out or melting of the compression member after the 100 hour run of the test fixture, it shall constitute failure of the test.

4.5.2.3.2.4 Size 3 and 4 belt performance test fixture. The size 3 and 4 belt performance test fixture shall consist of: an adjustable pulley spindle driven by a constant speed motor at 4900 rpm, ±75 rpm; an adjustable waterbrake pulley spindle, mechanically attached to a torque-type waterbrake; an adjustable tension pulley spindle; a means of tensioning the tension pulley spindle; a means of locking all adjustable pulleys in position; a means of aligning all pulleys in the same true plane; a set of test pulleys; and, a means of determining belt slip to an accuracy of plus or minus 1 percent. The size 3 and 4 belt performance test fixture configuration shall be as illustrated in figure 3. Test pulley sizes shall be as shown in the figure.

4.5.2.3.2.5 Size 3 and 4 belt performance test procedure. Mount the applicable size test pulleys on the spindles for the size of the test belt. Place the test belt on test pulleys. Adjust the test pulley spread until center distances and belt angles are as illustrated in figure 3. Apply and lock the applicable tension shown in the figure for the test belt size. Align the pulleys to run in the same plane. Turn the constant speed drive motor on. Do not apply waterbrake torque resistance. Allow the fixture to run for 5 min to precondition the test belt. If the test belt breaks or turns over in the test pulleys during preconditioning, it shall constitute failure. Allow the belt to cool for not less than 10 min, then retension and realign the fixture pulleys, if necessary. Turn the drive motor on again, and apply the applicable horse-power load (torque resistance by the waterbrake) for the test belt size as shown in the figure. Continue to operate the test fixture in this condition for 50 hours, while monitoring belt slippage. Verify the accuracy of the means to determine belt slippage with a calibrated, electronic digital counter at 10 hour intervals during the test. If slippage exceeds 8 percent at any one time during the test period, the drive motor may be stopped and the belt allowed to cool for 20 min, then pulley center distances, tension, and alignment may be restored and locked. The cooling and adjustment time shall not be counted as part of the test period. Restart the drive motor and restore the waterbrake load and continue the test and monitoring. At the end of the test period, stop the motor, release tension, remove the belt and examine for evidence of fatigue.



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4.5.2.3.2.6 Size 3 and 4 belt performance failure. If the test belt breaks or turns over in the pulleys at any time during the 50 hour run of the test fixture, it shall constitute failure of the test. If the belt slippage exceeds 8 percent at any time during the test period, after one adjustment of the fixture to compensate for initial slippage, it shall constitute failure of the test. If the belt exhibits evidence of fatigue, such as separation between rib and compression member in any degree, chipping of the rib, visible separation of a tension member, or chunking out or melting of the compression member after the 50 hour run of the test fixture, it shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for delivery shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

115. Materials, methods and containers not as specified.
116. Belts not preserved in sets or in the quantities specified.
117. Marking missing, illegible or incorrect.

5. PACKAGING

5.1 Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Belt sets. Belts provided in sets (see 3.9.2 and 6.2) shall be preserved in individual sets in accordance with MIL-P-116, Method III utilizing a PPP-B-636, W6c fiberboard box.

5.1.1.2 Single belts. Single belts of like description in the quantity specified (see 6.2) shall be preserved together as specified in 5.1.1.1.

5.1.2 Commercial. The belts shall be preserved in accordance with the applicable requirements of ASTM D 3951.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).



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5.2.1 Level A. The belts shall be packed in a wood box conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The belts shall be packed as specified in 5.2.1 except the boxes shall be domestic type.

5.2.3 Commercial. The belts shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for levels A and B shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial preservation and packing shall be in accordance with ASTM D 3951. In addition, the cube and weight shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The polyurethane V-belts are intended for use in power transmission applications that require high-speed, small-pulley, short-center drives, such as engine-driven alternator, pump, fan and similar accessories, in ambient temperature conditions as low as -54° C (-65° F) and as high as 82° C (180° F). The polyurethane metric belts are suitable for use in existing drives, as follows:

| Drives using these nominal belt sizes (width) | (thickness) | Use this size of polyurethane belt in equal effective length |
|---|---|---|
| 1/8 in. | 3/32 in. | 1 |
| 3/16 in | 1/8 in. | 2 |
| 5/16 in. | 7/32 in. | 3 |
| 7/16 in. | 9/32 in. | 4 |

Polyurethane V-belts exhibit a tendency to slip when moisture is present on the drive components. Their use in moisture-laden atmospheres, or in conditions wherein the drive may be exposed to moisture, requires consideration of this characteristic.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Specification part number (see 1.2.1).



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c. When first article is required for inspection and approval, and the number of units required (see 3.2).

d. When belts less than 12 months old are required (see 3.9.1).

e. When a matched set is required andthe quantity of belts in the set (see 3.9.2).

f. Degree of preservation and packing required (see 5.1 and 5.2).

g. Quantity of single belts to be preserved together for level A (see 5.1.1.2).

6.3 First article. When a first article inspection is required, the units should be samples selected from the contractor's current inventory. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and test approval of the document's first article.

Custodians:
Army - ME
Navy - YD
Air Force - 99

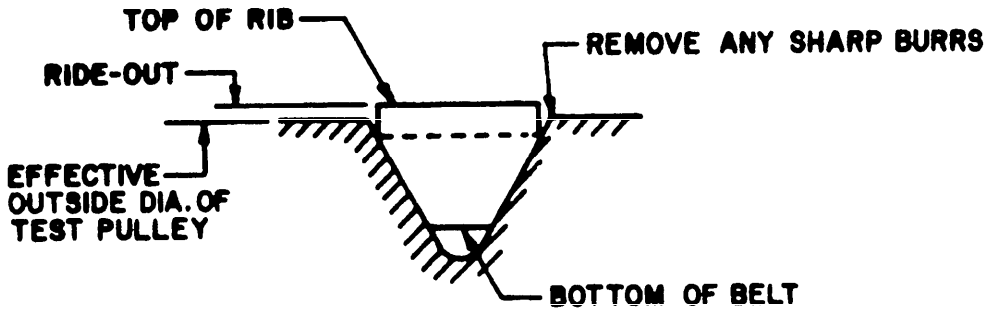
Review activities:
Air Force - 82

Preparing activity:
Army - ME

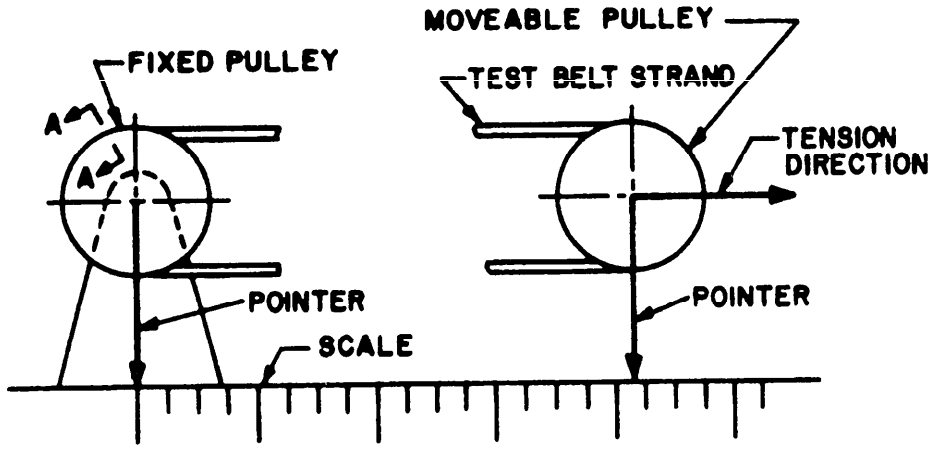
Project 3030-0154



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SECTION A-A




| TEST BELT SIZE | PULLEY OUTSIDE DIAMETER | | PULLEY GROOVE ANGLE | DEPTH OF GROOVE | |
|---|---|---|---|---|---|
| | mm | in. | degrees | mm | in. |
| 1 | 25.273 | .995 | 62 ± 1/4 | 1.880 | .074 |
| 2 | 80.823 | 3.183 | 62 ± 1/4 | 3.150 | .124 |
| 3 | 80.823 | 3.183 | 60 ± 1/4 | 5.283 | .208 |
| 4 | 80.823 | 3.183 | 60 ± 1/4 | 8.255 | .335 |

TEST PULLEY DIMENSIONS

| BELT SIZE | TEST TENSION | |
|---|---|---|
| | kilograms | pounds |
| 1 | 13.61 | 30 |
| 2 | 27.22 | 60 |
| 3 | 54.43 | 120 |
| 4 | 90.72 | 200 |

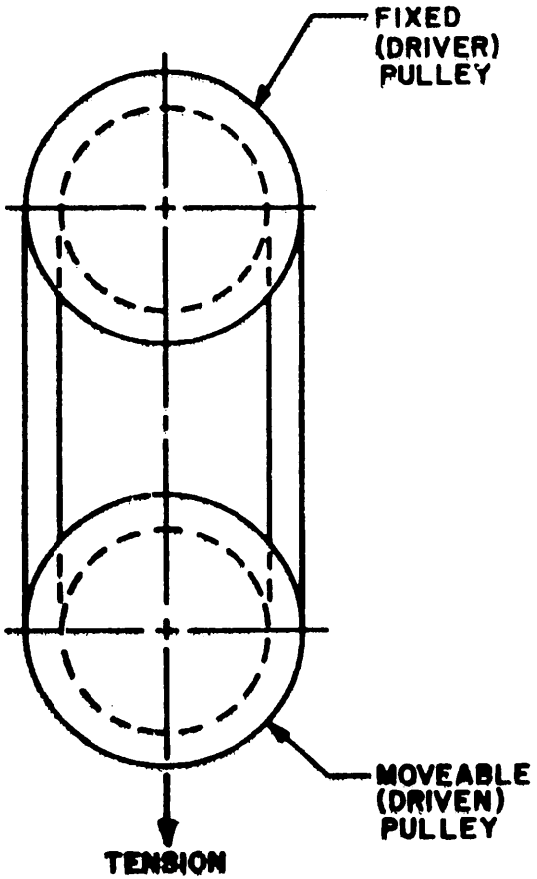
TEST TENSIONS

FIGURE 1. Dimensional test fixture.

X-4308



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| TEST BELT SIZE | PULLEY OUTSIDE DIAMETER | | PULLEY GROOVE ANGLE | PULLEY GROOVE | | | |
|---|---|---|---|---|---|---|---|
| | | | | TOP WIDTH | | DEPTH | |
| | mm | in. | degrees | mm | in. | mm | in. |
| 1 | 12.700 | .500 | 60 ± 1/4 | 2.794 | .110 | 1.88 | .074 |
| 2 | 25.400 | 1.000 | 60 ± 1/4 | 2.972 | .117 | 3.150 | .124 |

TEST PULLEY DIMENSIONS

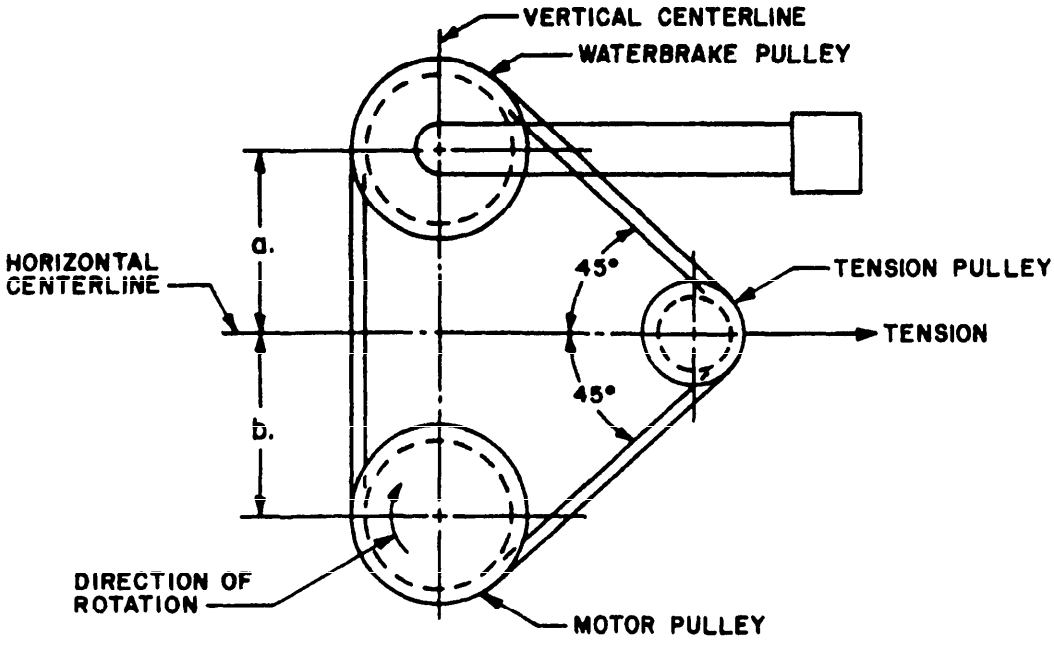
| BELT SIZE | TEST TENSION | |
|---|---|---|
| | kilograms | pounds |
| 1 | 11.34 | 25 |
| 2 | 36.29 | 80 |

TEST TENSIONS

FIGURE 2. Size 1 and 2 belt performance test fixture.

X-4309

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| TEST BELT SIZE | PULLEY OUTSIDE DIAMETER | | PULLEY GROOVE ANGLE | PULLEY GROOVE | | | |
|---|---|---|---|---|---|---|---|
| | | | | TOP WIDTH | | DEPTH | |
| | mm | in. | degrees | mm | in. | mm | in. |
| 3 | 120.650 | 4.750 | 62 ±1/4 | 7.112 | .280 | 5.283 | .208 |
| 4 | 120.650 | 4.750 | 62 ±1/4 | 10.439 | .441 | 8.509 | .335 |

MOTOR AND WATERBRAKE PULLEY DIMENSIONS

| TEST BELT SIZE | PULLEY OUTSIDE DIAMETER | | PULLEY GROOVE ANGLE | PULLEY GROOVE | | | |
|---|---|---|---|---|---|---|---|
| | | | | TOP WIDTH | | DEPTH | |
| | mm | in. | degrees | mm | in. | mm | in. |
| 3 | 42.875 | 1.688 | 60 ±1/4 | 7.112 | .280 | 5.588 | .220 |
| 4 | 47.625 | 1.875 | 60 ±1/4 | 10.439 | .441 | 8.534 | .336 |

TENSION PULLEY DIMENSIONS

| BELT SIZE | TEST TENSION | |
|---|---|---|
| | kilograms | pounds |
| 3 | 60.78 | 134 |
| 4 | 61.23 | 135 |

TEST TENSIONS

| BELT SIZE | WATERBRAKE LOAD horsepower |
|---|---|
| 3 | 13.4 |
| 4 | 14.5 |

WATERBRAKE LOADS

NOTE: UPPER DIMENSION "a." SHALL BE EQUAL TO LOWER DIMENSION "b.".

FIGURE 3. Size 3 and 4 belt performance test fixture.

X-4310



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| DOD-B-53061 | Belts, V: Polyurethane |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

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